

Assembly-Induced Membrane Selectivity of Artificial Model Peptides through Entropy–Enthalpy Competition

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ABSTRACT: Peptide design and drug development offer a promising solution for combating serious diseases or infections. In this study, using an AI–human negotiation approach, we have designed a class of minimal model peptides against tuberculosis (TB), among which K7W6 exhibits potent efficacy attributed to its assembly-induced function. Comprising lysine and tryptophan with an amphiphilic α -helical structure, the K7W6 sequence exhibits robust activity against various infectious bacteria causing TB (including clinically isolated and drug-resistant strains) both *in vitro* and *in vivo*. Moreover, it synergistically enhances the effectiveness of the first-line antibiotic rifampicin while displaying low potential for inducing drug resistance and minimal toxicity toward mammalian cells. Biophysical experiments and simulations elucidate that K7W6's exceptional performance can be ascribed to its highly selective and efficient membrane permeabilization activity induced by its distinctive self-assembly behavior. Additionally, these assemblies regulate the interplay between enthalpy and entropy during K7W6–membrane interaction, leading to the peptide's two-step mechanism of membrane interaction. These findings provide valuable insights into rational design principles for developing advanced peptide-based drugs while uncovering the functional role played by assembly.

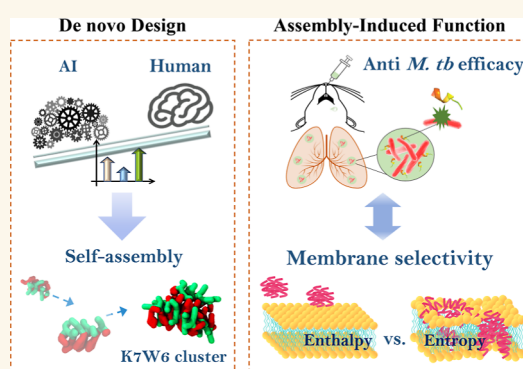
KEYWORDS: peptide design, assembly-induced function, membrane selectivity, entropy–enthalpy competition, peptide–lipid interaction

INTRODUCTION

The difficult-to-treat pathogen infections pose a significant threat to global health.¹ Among these, tuberculosis (TB), particularly TB caused by multidrug-resistant *Mycobacterium tuberculosis* (*M. tb*; the causative agent of TB²), stands out as one of the most highly contagious and lethal diseases.^{3,4} In recent years, there have been 8.7 million newly increased cases of TB annually, resulting in 1.5 million deaths, thus emphasizing the urgent necessity for the discovery of effective drugs.^{5,6}

The development of peptide-based drugs offers a solution to this critical issue by harnessing the vast diversity of amino acid sequences, which endows peptides with exceptional structural and functional versatility.^{7–10} Consequently, they hold great promise as an invaluable resource for drug discovery and development.^{11–13} However, fully exploring the design space of peptides seems impossible due to its extremely high dimension ($\sim 20^n$), where n represents the length of the peptide chain and 20 denotes the number of commonly available amino acid types (Figure 1a). For instance, even short

peptides with $n = 10$ can have over 1 trillion possible sequences. Recently, artificial intelligence (AI) and machine learning have emerged as powerful strategies to overcome this challenge while effectively balancing exploration-versus-exploitation trade-offs.^{7–10,14,15} By employing a machine-learning pipeline following a coarse-to-fine design principle, Huang *et al.* successfully identified three antimicrobial hexapeptides from billions of potential peptide sequences that exhibited potent activities against multidrug-resistant pathogens.⁷ Nevertheless, de novo peptide design still poses numerous challenges and obstacles for machine learning. As a data-driven methodology, its outcomes would be impaired by the limited number of available studies or inconsistent data sets resulting from

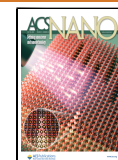


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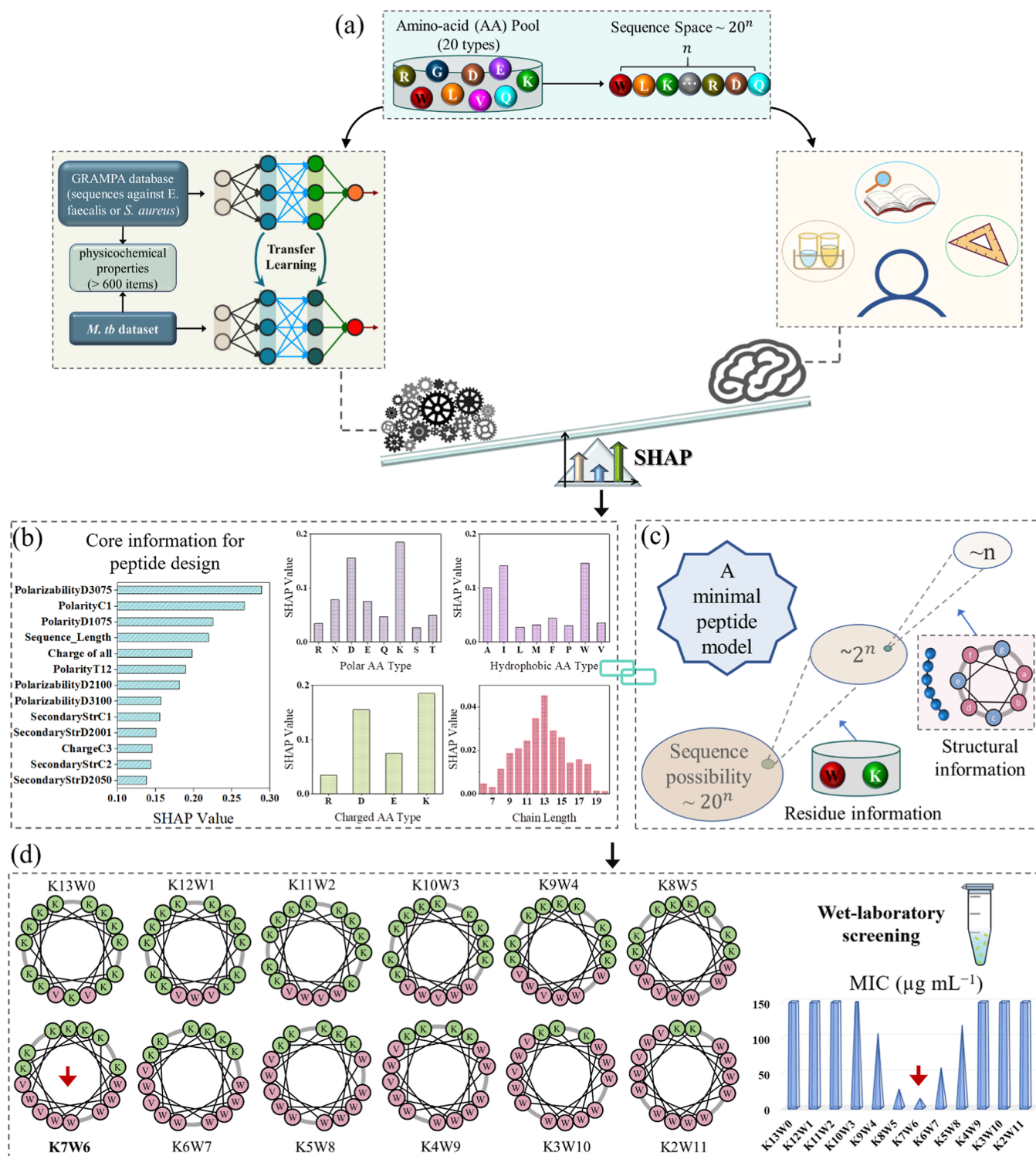


Figure 1. “AI–human negotiation” framework for peptide design. (a) Illustrations of the machine learning model and the dialogue mode between AI expert and human. (b) Encoding of key influencing factors in peptide sequences with predicted antimycobacterial activities using the machine-learning model based on *E. faecalis* and *M. tuberculosis* (*M. tb*) data sets, along with SHAP analysis results. (c) Dimension reduction sketch of the peptide design space incorporating additional information. (d) Design progression from K13W0 to K2W11, accompanied by helical wheel diagrams illustrating residue distribution, as well as wet-laboratory validation of biological activities through minimum inhibitory concentration (MIC) tests against *Mycobacterium smegmatis* mc²155.

experimental variations.^{9,10} Additionally, different peptides may possess distinct working mechanisms that are not directly incorporated into existing peptide databases, further impacting the efficacy of machine learning algorithms.¹⁶ In fact, in nearly all machine-learning scenarios, peptides with higher predicted

scores may exhibit poorer biological activity in actual tests.^{7–10,15,17} On the other hand, human expertise has been proven helpful in improving peptide performance despite potential biases present within it.^{8,18–20} Therefore, effectively leveraging the advantages of machine learning to augment

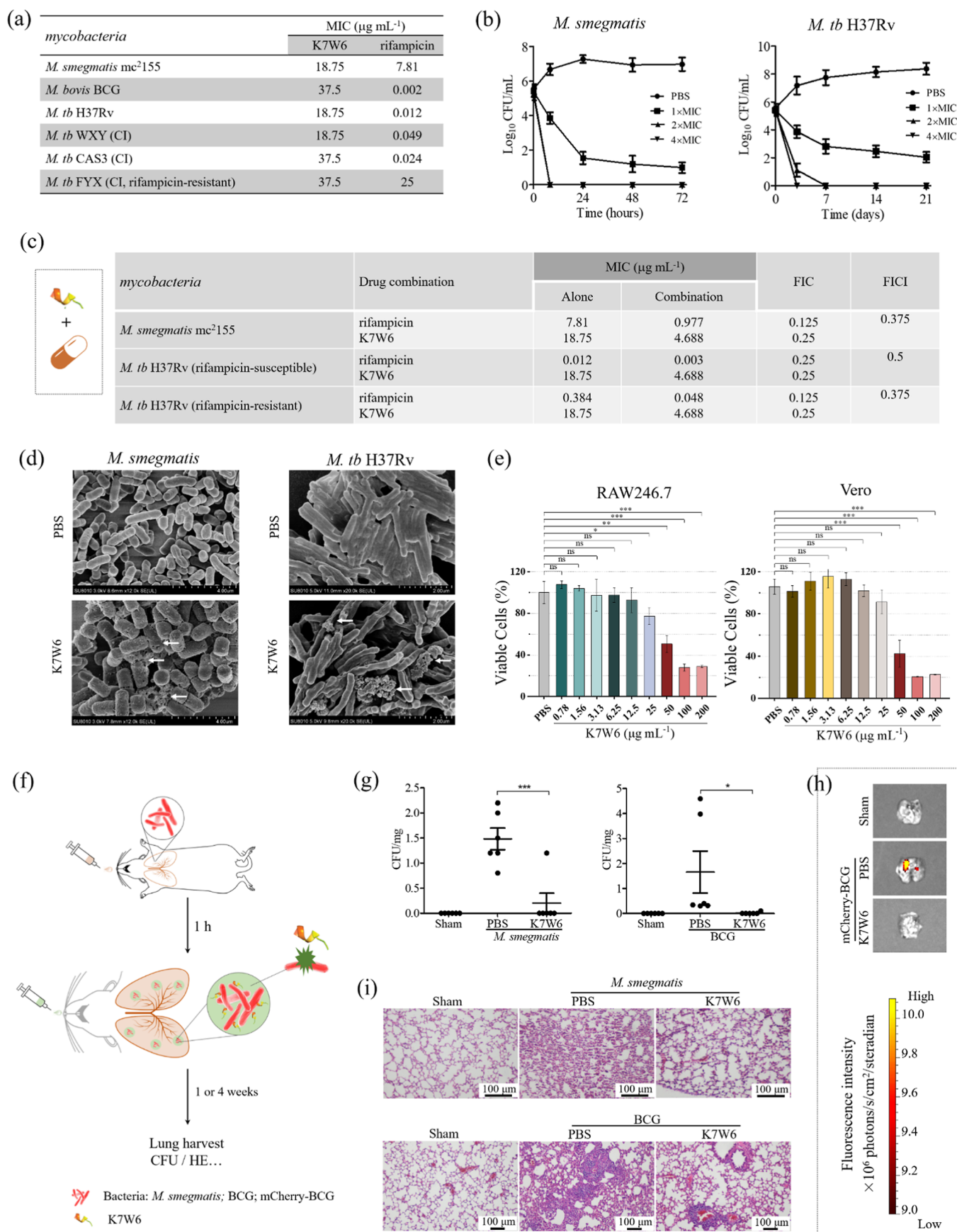


Figure 2. Antimycobacterial and cytotoxic activities of K7W6. (a) *In vitro* antimycobacterial activity of K7W6. Rifampicin was used as a positive control. CI, clinically isolated. (b) Killing curves for different concentrations (1 $\times$, 2 $\times$, or 4 $\times$ MIC) of K7W6 after incubation for various durations against *M. smegmatis* or *M. tb* H37Rv. (c) Checkerboard assay showing the synergistic actions between K7W6 and

Figure 2. continued

rifampicin. FIC, fractional inhibitory concentration; FICI, fractional inhibitory concentration index. The data in (a,c) are mean values representing three independent experiments performed in duplicates. (d) SEM micrographs revealing morphological changes and membrane disruptions in mycobacteria (representatively marked by arrows) treated with K7W6 at 1× MIC for 4 h. PBS was used as the control. (e) Cytotoxic assays demonstrating the effects of K7W6 on RAW264.7 and Vero cells after peptide incubation for 24 h. The experiments in (b,e) are performed as three biologically independent experiments, and the mean $\pm$ SD is shown. (f–i) Activity of K7W6 against *M. tb* infection in C57BL/6J mice. (f) Schematic diagram illustrating the *in vivo* experimental protocol. C57BL/6J mice (18–20 g, $n = 6$) were intratracheally infected with *M. smegmatis*, *M. bovis* BCG, or mCherry-*M. bovis* BCG. 1 h postinfection, K7W6 (5 mg kg⁻¹) or PBS (the solvent of K7W6) was administered into the mouse airways using a pipette tip. Sham mice received an equivalent volume of vehicle (PBS). 1 week after inoculation with *M. smegmatis* and 4 weeks after inoculation with *M. bovis* BCG, the mice were euthanized by cervical dislocation for subsequent evaluation of colony-forming units (CFUs) counts, fluorescence intensity, and lung histopathology analysis. (g) Quantification of viable CFUs in the lung. (h) Fluorescent imaging depicting mCherry-*M. bovis* BCG distribution in the lung. (i) Histopathological examination of the lung tissue. * $p < 0.05$ and *** $p < 0.001$. Error bars represent standard error.

human knowledge holds greater promise for advancing peptide-based drug development.

The activities of peptides involve intricate mechanisms, such as supramolecular assembly, which are also crucial for peptide engineering in the development of therapeutic agents and bioimaging probes with improved performance.^{21–29} Sun *et al.* discovered that self-assembling tripeptides (Phe–Phe–Tyr) effectively reverse drug resistance in melanoma by interfering with microtubules.³⁰ Furthermore, the experiments demonstrated that peptide self-assembly into micelles followed by transformation into nanofibers can induce apoptosis in cancer cells by disrupting HER2 dimerization.³¹ However, our current understanding of manipulating peptide self-assembly to achieve specific functionalities remains inadequate.

In this study, we employed an “AI–human negotiation” framework with Shapley additive explanation (SHAP) analysis to develop a minimal model peptide sequence against TB (Figure 1). Consequently, a series of helical sequences composed of lysine (K) and tryptophan (W) residues with varying hydrophobicity and charged states were generated, ranging from K13W0 to K2W11. Among them, the K7W6 sequence exhibited the most potent antimycobacterial activity both *in vitro* and *in vivo* while demonstrating low cytotoxicity. Furthermore, it synergized effectively with the first-line antibiotic rifampicin. Through biophysical assays, it was found that the K7W6 sequence displayed distinct membrane activities depending on its aggregation state: self-assembly of K7W6 triggered enthalpy-controlled selective binding to specific membranes followed by entropy-driven disassembly of peptides into monomers, fully exploiting the superior properties of individual sequences. Overall, our findings provide valuable insights for rational machine-assisted peptide design against TB while highlighting unexpected functions induced by assembly in highly selective and efficient membrane activity.

RESULTS AND DISCUSSION

Machine-Assisted Peptide Design and Training of Classification Models. The utilization of machine learning enables a comprehensive exploration of the vast peptide design space, effectively reducing dimensions for specific objectives. The GRAMPA (Giant Repository of AMP Activities) database, which encompasses over 51,000 laboratory results on peptides and their corresponding minimum inhibitory concentration (MIC) values,^{32,33} was selected for the machine learning model training. However, due to limited data (~50 records) available on *M. tb*, a strategy combining deep learning and transfer learning was employed (Figure 1a). Specifically, data sets from

Enterococcus faecalis were chosen as the foundation for constructing a multilayer perceptron (MLP) model for peptide classification based on correlation analysis of antibacterial activities (Figure S1). Physicochemical descriptors⁷ were assigned to represent peptide sequences as structured data points (~677 items per data point) for input purposes. The resulting pretrained model was then fine-tuned specifically using *M. tb* data sets and demonstrated exceptional performance (for more details, refer to the Supporting Information).

In principle, the trained machine-learning model can be directly applied for predicting antimycobacterial peptide sequences. However, there are inherent risks associated with this prediction.⁷ On the other hand, the main challenge in human peptide design lies in the vast sequence space and limited understanding of the corresponding sequence–function relationship. Nevertheless, this model can aid in comprehending these aspects and especially assist us in developing a “minimal model peptide against TB”, which is the specific objective of this study. Here, the term “minimal model peptide” refers to an amino-acid sequence that possesses minimal required types, appropriate chain length, and secondary structure. Therefore, we engaged in a dialogue with the “AI expert” to gain a better understanding of the peptide design space through SHAP analysis.³³ The key information provided by the machine-learning model for such peptide design is presented in Figure 1b and Table S1. Analysis of top-ranked items revealed that amino acid properties such as polarity, amphipathicity, and charge state play significant roles, suggesting that at least two types of amino acids are necessary to meet our design requirements. Accordingly, polar lysine (K) with positive charge and hydrophobic tryptophan (W) were recommended by the model (Figure 1b), reducing the dimensionality from 20ⁿ to 2ⁿ (Figure 1c). Considering structural properties like chain length and secondary structure (Figure 1b and Table S1), a sequence length of $n = 13$ was suggested within a range from $n = 5–20$. Moreover, an amphipathic α -helical structure holds significance,^{34–36} which further narrows down the possible sequences to approximately 13 (Figure 1c,d). Interestingly, our transfer learning model trained on the *Staphylococcus aureus* data set yielded nearly identical results compared to *M. tb* antibacterial activities (Figure S2a) while being distinct from the outcomes obtained through direct database statistics (Figure S2b).

The designed peptide sequences were synthesized and evaluated for their antimicrobial efficacy in the final screening phase (K1W12 was excluded due to its pronounced hydrophobicity; please refer to corresponding note to Table S2). As illustrated in Figure 1d and Table S2, these sequences

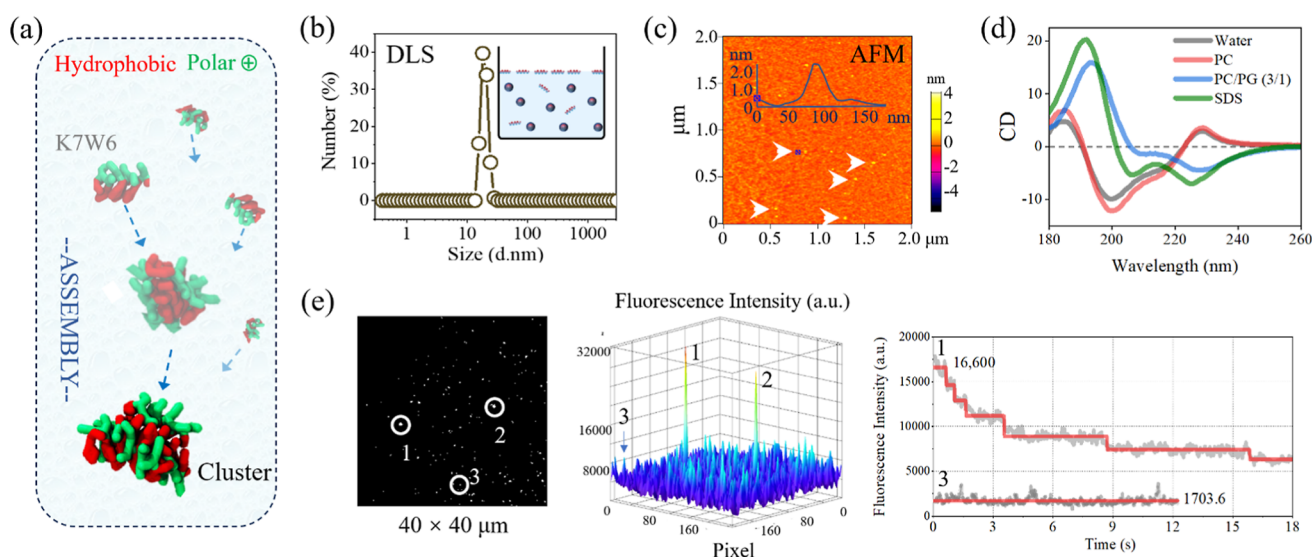


Figure 3. Characterization of self-assembled K7W6 clusters. (a) Schematic illustrating the self-assembly of K7W6 into clusters. (b–e) Dynamic light scattering (DLS, b), atomic force microscopy (AFM, c), circular dichroism (CD, d), and single-molecule stepwise photobleaching (e) characterizations of K7W6 (at $10.0 \mu\text{g mL}^{-1}$). The inset in (b) depicts the formation of K7W6 clusters in solution above the threshold concentration. The white spots in (c), indicated by white arrowheads, denote self-assembled clusters, with one of the clusters typically presented with a height profile. The profiles in (d) show peptides dissolved in aqueous solutions containing different molecules such as pure phosphatidylcholine (PC; at $250 \mu\text{g mL}^{-1}$), PC/phosphatidylglycerol (PG) mixture (3/1 by mol; lipid = $250 \mu\text{g mL}^{-1}$), or sodium dodecyl sulfate (SDS; at 60 mM). The image in (e) is captured using rhodamine B-labeled K7W6 under total internal reflection fluorescence microscope (TIRFM). Spots 1 and 2 represent peptide clusters, while spot 3 represent a peptide monomer (*i.e.*, a single fluorescent dye).

exhibited varying levels of antimycobacterial activity attributed to differences in hydrophobicity and charge states. Among them, K7W6 (composed of 7 lysine and 6 tryptophan residues) demonstrated the most potent antimycobacterial activity with a MIC value of $6.3 \mu\text{M}$ (equivalent to $13.9 \mu\text{g mL}^{-1}$) against *Mycobacterium smegmatis* mc²155 during live cell assays. Overall, the successful generation of a peptide sequence with potent antimycobacterial properties can be attributed to the effective integration of an “AI–human negotiation” framework and the design concept of “minimal peptides”.

Potent Antimycobacterial Activity of K7W6 In Vitro and In Vivo. The efficacy of K7W6 against TB was thoroughly assessed using live cell assays. As depicted in Figure 2a, K7W6 exhibited potent activity against *M. smegmatis* mc²155, *M. tb* H37Rv, and *Mycobacterium bovis* BCG, with MIC values of 18.8 or $37.5 \mu\text{g mL}^{-1}$, respectively; moreover, it demonstrated effectiveness against clinically isolated *M. tb* strains and rifampicin-resistant *M. tb* strains, with MIC values ranging from 18.8 to $37.5 \mu\text{g mL}^{-1}$. The bactericidal ability of K7W6 was also confirmed through killing kinetic assays at concentrations equivalent to 1 $\times$, 2 $\times$, and 4 $\times$ the MIC against both *M. smegmatis* and *M. tb* H37Rv (Figure 2b). Additionally, combination drug regimens play a crucial role in the success of anti-TB chemotherapy; henceforth, the synergistic effect between K7W6 and rifampicin—the most effective first-line antibiotic for TB treatment—was evaluated using checkerboard assay.³ Notably, K7W6 displayed strong synergy with rifampicin against both *M. smegmatis* strains as well as rifampicin-susceptible/resistant variants of *M. tb* H37Rv (Figure 2c). Furthermore, drug resistance tests were conducted, which revealed that repeated exposure to K7W6 for up to 10 cycles did not lead to an increase in the MIC value (Table S3a); this suggests that bacteria do not readily develop

resistance toward this drug. In contrast, under identical conditions, the MIC value for rifampicin increased by either a factor of 32 or 64; however, K7W6 effectively reduces this factor to 4 or 8 (Table S3b and corresponding note). Scanning electron microscopy (SEM) observation (Figure 2d) as well as extracellular ATP assays (Figure S3) clearly indicated that the action of K7W6 on *M. tb* disrupted cell membranes significantly, suggesting a potential antibacterial mechanism.

On the contrary, K7W6 demonstrated moderate cytotoxicity against mammalian cells, resulting in a cellular mortality rate of less than 20% at a concentration of $25 \mu\text{g mL}^{-1}$ (comparable to the MIC value) after 24 h of incubation. Even at concentrations up to $200 \mu\text{g mL}^{-1}$, K7W6 did not completely inhibit the growth of RAW246.7 and Vero cells (Figure 2e). These observations suggest that K7W6 exhibits selective action against bacteria while maintaining favorable biocompatibility with mammalian cells.

The antimycobacterial activities of K7W6 were further evaluated *in vivo* using C57BL/6J mice. To accomplish this, we infected the mice with *M. smegmatis*, *M. bovis* BCG, or mCherry-*M. bovis* BCG via intratracheal instillation; subsequently, K7W6 or its solvent (PBS) was administered via intratracheal instillation as nasal drops to ensure a higher local efficacy concentration for respiratory tract infections (Figure 2f). Our results clearly demonstrated that treatment with K7W6 (5 mg kg^{-1}) significantly reduced the colony-forming units (CFUs) of *M. smegmatis* and *M. bovis* BCG in the lungs compared to the PBS-treated group (Figure 2g). Furthermore, treatment with K7W6 led to a noticeable decrease in fluorescence intensity emitted by mCherry-*M. bovis* BCG within the lungs of mice (Figure 2h). Additionally, we observed that treatment with K7W6 (5 mg kg^{-1}) effectively alleviated the pathological changes occurring in mouse lungs upon infection with *M. smegmatis* and *M. bovis* BCG strains

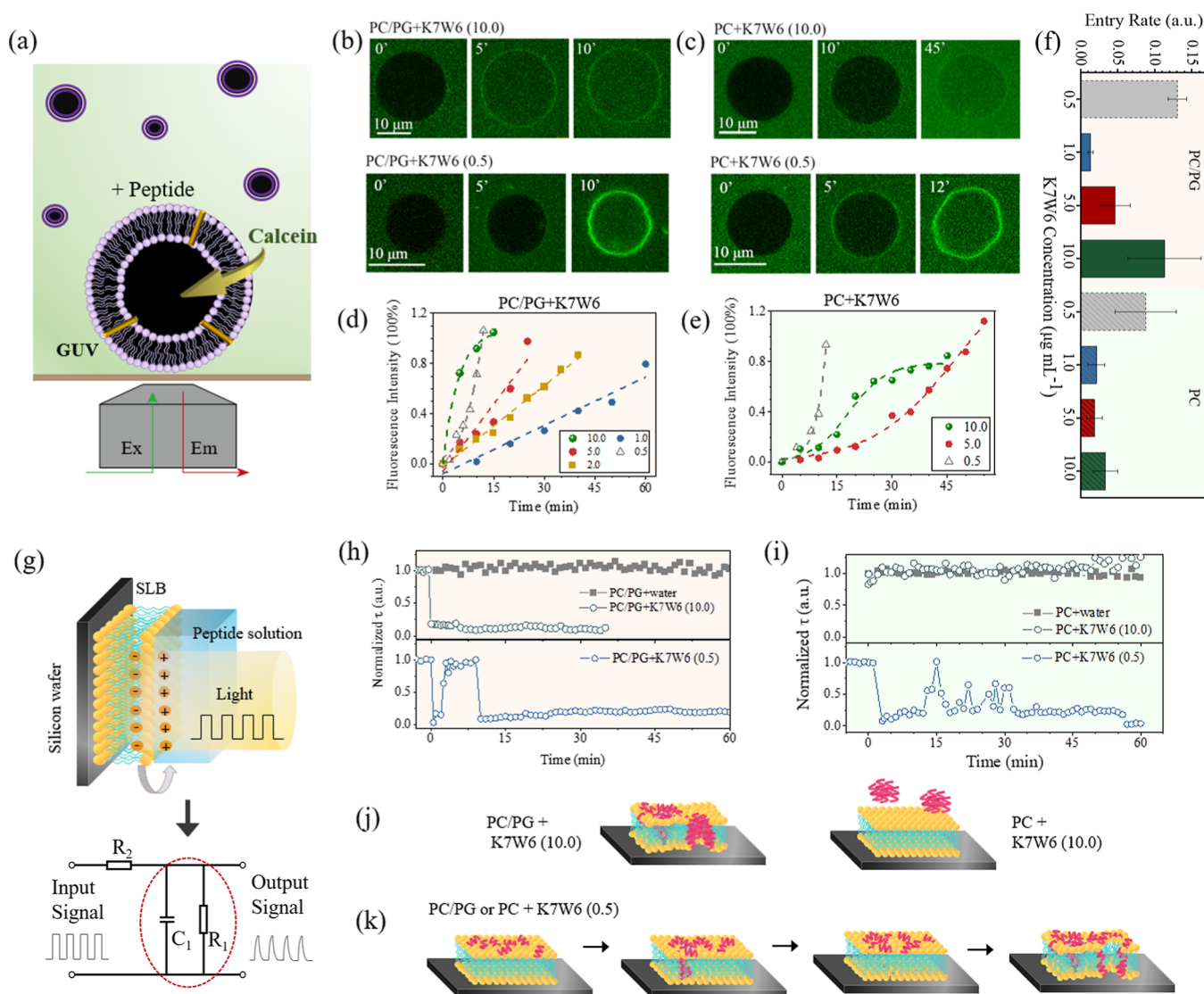


Figure 4. Distinct action dynamics of K7W6 over PC and PC/PG membranes at various concentrations. (a–f) Dynamic GUV leakage assays, including the schematic diagram (a), representative confocal images (b,c), corresponding time evolution of fluorescence intensity of calcein inside individual GUVs after normalization (d,e), and histograms of mean entry rate of calcein into GUVs after K7W6 addition at different concentrations (f). (g–k) Photovoltage transient monitoring, including cartoons illustrating setup and equivalent circuit of the test system (g), representative time evolution of τ of a PC or PC/PG membrane after K7W6 exposure at different concentrations (h,i), and schematic diagrams showing possible structural disturbances caused by peptides (as clusters or monomers) on the membrane (j,k). The membranes are composed of pure PC or PC/PG (9/1 by mol) and peptides were added at time point 0 min. Representative GUVs are shown in (b–e) and representative SLBs are shown in (h,i). Images in (b,c) were captured in the green channel for calcein and red channel for lipids, although only the green channel images are shown here to conserve space. The corresponding concentration of peptides (in $\mu\text{g mL}^{-1}$) and time (in min) for each image are noted in the images. The scatter diagrams in (d,e) are fitted using a linear, $y = kx + b$, or a sigmoidal function, $y = a/(1 + e^{-b(t-t_c)})$ (k , a , b , and t_c are constants). Data in (f) are shown as the mean of at least five replicas $\pm$ standard error of the mean. (g) Setup of the working electrode and the electric double layer at the interface between the bilayer membrane and aqueous solution; the variable C_1 and R_1 in parallel refer to the contribution of the working electrode, and the constant R_2 corresponds to the impedance of the rest circuit. The output voltage signal is collected simultaneously. In (h,i), the condition of adding solvent (water) is used as a control. In (j,k), the hydrophilic headgroups and hydrophobic alkyl tails of phospholipids are shown as yellow beads and light cyan lines, respectively. Red coils denote the peptide.

(Figure 2i). Overall, these findings provide evidence for the effective role played by K7W6 in restraining *M. tb* infection within an *in vivo* setting.

Self-Assembly and the Resulting Functionality of K7W6 Sequences. The functions of K7W6 are correlated with its physicochemical properties and action mechanism. In experimental observations, K7W6 undergoes self-assembly into clusters in solution (Figure 3a). The formation of stable peptide clusters occurs at concentrations equal to or exceeding

$10.0 \mu\text{g mL}^{-1}$, with an average size of approximately 20 nm, as characterized by both dynamic light scattering (DLS) and atomic force microscopy (AFM) techniques (Figure 3b,c). This phenomenon can be attributed to the balance between hydrophobic and polar/charged residues in its sequence, resulting in an amphipathic property with a helical structure (Figures 1d, 3d and S4—corresponding note). At lower concentrations (e.g., 5.0, 2.5, or $1.3 \mu\text{g mL}^{-1}$), a dynamic assembly disassembly process is observed during real-time DLS

measurements, while minimal aggregation is detected at concentrations below or equal to $0.6 \mu\text{g mL}^{-1}$ under either DLS or AFM analysis, indicating the presence of monomeric peptides (Figure S4 and corresponding note; note that a reading exceeding $1 \mu\text{m}$ under DLS indicates the absence of distinguishable stable particles in the system due to measurement limitations). Further characterization using single-molecule stepwise photobleaching confirms the formation of clusters consisting of around 8–10 peptides each with a size of approximately 20 nm (Figures 3e and S4d and corresponding note). These findings suggest that K7W6 peptides can undergo concentration-dependent self-assembly into stable clusters, which may be associated with their potent antibacterial activity. Concentrations of 10.0 and $0.5 \mu\text{g mL}^{-1}$ were chosen as representative concentrations for subsequent biophysical experiments to represent the cluster and monomer states, respectively.

For cationic peptides, membrane disruption or perturbation is one of the primary mechanisms underlying their biological activities.^{34,37} Moreover, our SEM and extracellular ATP assays (Figures 2d and S3) have confirmed the significant bacterial membrane disruption caused by K7W6. Therefore, biophysical assays with model systems were conducted to meticulously investigate the dynamic mechanism of action on cell membranes.^{38,39} First, we employed a dynamic giant unilamellar vesicle (GUV) leakage assay^{40,41} to assess the membrane permeabilization ability of K7W6 at various concentrations (ranging from 10.0 to $0.5 \mu\text{g mL}^{-1}$; Figure 4a). GUVs composed of pure phosphatidylcholine (PC) lipids or a mixture of PC and phosphatidylglycerol (PG) lipids in a 9:1 molar ratio, mimicking mammalian and bacterial cell membranes, respectively,⁴¹ were prepared through electroformation. The GUVs dispersed in calcein solution remained stable for over 4 h without deformation or calcein entry under confocal observation. However, upon addition of K7W6 at a concentration of $10.0 \mu\text{g mL}^{-1}$, significant transmembrane influx of calcein into the vesicles was observed (Figures 4b,c and S5). Notably, the entry rate of calcein was considerably faster for the PC/PG vesicles, completing within 5–10 min after peptide addition, compared to the PC vesicles, which took around 45 min. To quantitatively assess the extent of membrane permeabilization, we analyzed the temporal evolution of fluorescent intensity within each GUV (Figure 4d,e) and calculated the stable entry rate k by fitting the linear stage of each profile (Figures 4f and S5g for details). Interestingly, despite employing an identical peptide concentration, we found that the entry rate for PC/PG membranes was over twice as high as that for PC membranes. These findings suggest that PG enhances the activity of K7W6 and may have potential for targeting bacterial membranes instead of mammalian cell membranes.

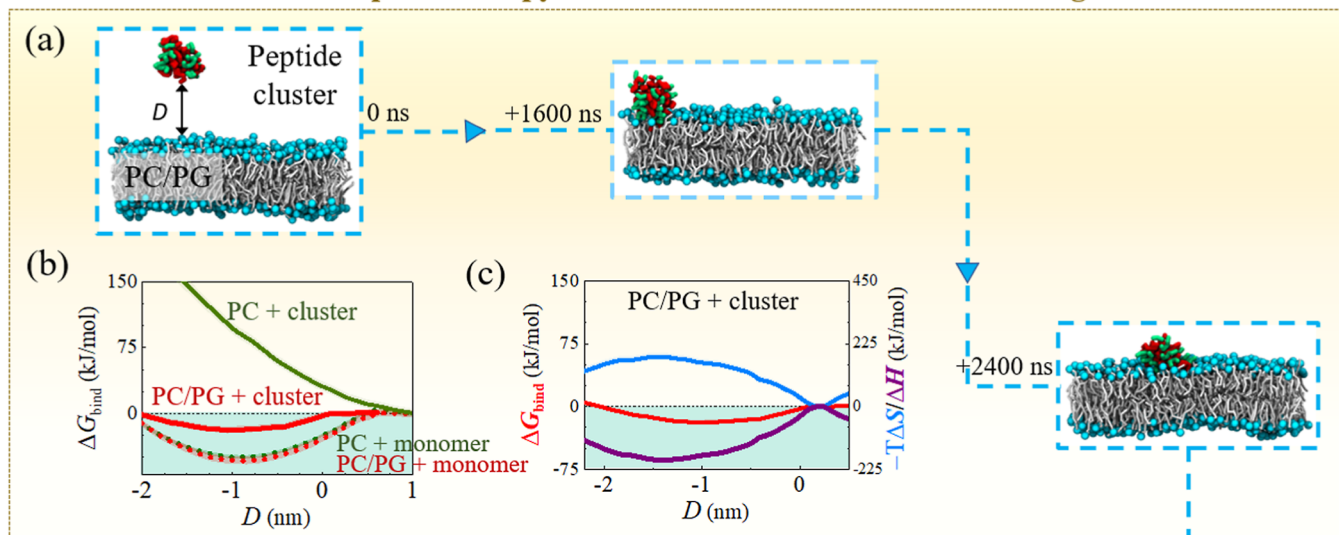
The concentration and permeabilization efficiency of most natural antimicrobial peptides usually exhibit a positive correlation,^{40,42,43} but this designed peptide violates this trend (Figures 4b,c and S5). Furthermore, we observed that the membrane specificity observed at a concentration of $10.0 \mu\text{g mL}^{-1}$ was significantly diminished at a low concentration of $0.5 \mu\text{g mL}^{-1}$. At this lower concentration, membrane-permeabilization effects were detected approximately 10 min after the addition of K7W6 in both PC and PC/PG systems. The presence of a bright green circle overlaid with the lipid membrane indicated the accumulation of calcein molecules within the membrane defects (please refer to the correspond-

ing note to Figure S5 for details). Additionally, the dynamic entry profiles for both PC and PC/PG membranes at a concentration of $0.5 \mu\text{g mL}^{-1}$ (represented by black triangles in Figure 4d,e) exhibited a sigmoidal shape (either whole or partial as estimated by fitting), indicating an accumulation-enhanced permeabilization effect of peptides. Conversely, the profiles corresponding to a concentration of $10.0 \mu\text{g mL}^{-1}$ peptide against a PC/PG membrane demonstrated an exponential shape (green beads in Figure 4d), suggesting immediate permeabilization upon peptide addition. These results highlight the concentration-dependent membrane activity of K7W6. In brief, the K7W6 monomers induced nonspecific perturbation of membranes; however, the peptide demonstrated assembly-triggered membrane selectivity with enhanced activity (at $10.0 \mu\text{g mL}^{-1}$) and exhibited significant bactericidal effects with minimal toxicity against mammalian cells in live cell assays (above MIC).

The membrane activity of K7W6 was further investigated using photovoltage transient testing, a recently developed method capable of sensitively detecting real-time changes in the membrane structure, particularly the formation of transmembrane defects.^{44–46} The working electrode consists of a silicon wafer substrate and a supported lipid bilayer (SLB) membrane, represented by a variable capacitor C_1 and a variable resistor R_1 in parallel in the simplified circuit (Figure 4g). When applying square-wave modulated light (e.g., 250 ms ON and 500 ms OFF), it generates a square-wave voltage signal due to the photoelectric response of the silicon substrate. The charge and discharge process of the SLB is simultaneously recorded as deformed square-wave pulses that depend on membrane properties. The voltage pulse can be quantitatively characterized as the relaxation time constant τ .⁴⁴ Thus, the real-time variation of membrane properties due to peptide actions is reflected by the time evolution of τ after adding peptides into solution.

The SLBs were prepared using the conventional vesicle-fusion method, consisting of PC/PG (at a 9:1 molar ratio) or pure PC. After system stabilization, K7W6 was added at various concentrations at 0 min. An immediate drop in τ to zero was observed upon the addition of peptides at a concentration of $10.0 \mu\text{g mL}^{-1}$ into the PC/PG membrane system, indicating highly efficient membrane permeabilization and instantaneous destruction (Figure 4h). In contrast, approximately half of the PC membranes remained largely unchanged within 15 min (Figure 4i), confirming the selective nature of K7W6 clusters toward membranes. However, when K7W6 was added at a concentration of $0.5 \mu\text{g mL}^{-1}$, different situations were observed. For the PC/PG membrane, addition of K7W6 caused fluctuations in τ within 10 min; thereafter, τ rapidly decreased and stabilized. These changes indicate that peptides at a concentration of $0.5 \mu\text{g mL}^{-1}$ immediately induced transmembrane defects in the PC/PG membrane (<1 min), which gradually progressed until complete permeabilization occurred (~ 10 min; Figure 4k). A similar but more complex change induced by K7W6 action was observed for the PC membrane (Figures 4i and S6): τ frequently fluctuated within 40 min before stabilizing and dropping to ~ 0 after 55 min. Peptides at other concentrations also showed distinct membrane action processes (Figure S7). In summary, at a low concentration of $0.5 \mu\text{g mL}^{-1}$, K7W6 exhibited a process involving permeation–healing–permeation on both membranes, while at a high concentration of $10 \mu\text{g mL}^{-1}$, it demonstrated selective and efficient breaking activity on PC/

Step 1: Enthalpy-controlled selective membrane binding



Step 2: Entropy-driven disassembly on membrane

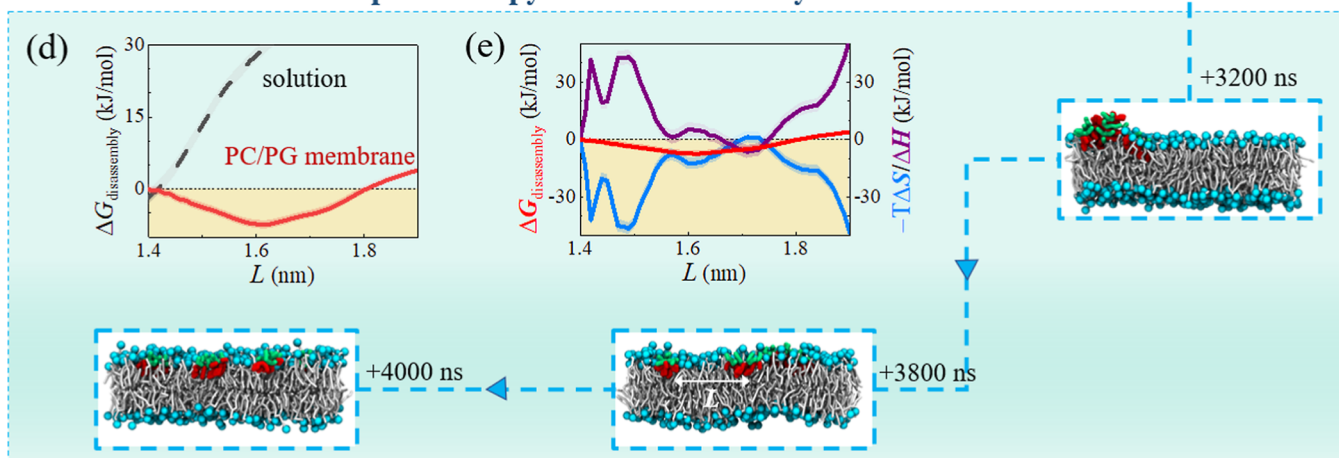


Figure 5. Two-step interaction mechanism between self-assembled K7W6 clusters and membranes. (a) Simulation snapshots depicting the binding process and subsequent disassembly of a peptide cluster composed of four K7W6 sequences on a PC/PG membrane. Red: tryptophan (W); green: lysine (K); cyan: lipid head; gray: lipid tail. (b,d) Free energy profiles for the binding of peptide monomers or clusters to various membrane systems (b), as well as for the disassembly of a peptide cluster on a PC/PG membrane or in solution (d). In these calculations, disassembly refers to the detachment of one peptide from the cluster. (c,e) Changes in enthalpy and entropy during membrane-binding (c) and disassembly processes (e).

PG membranes. More importantly, Figures 3 and S4 reveal that K7W6 exists as monomers and clusters at these two concentration conditions. Therefore, these biophysical experiments unveil a compelling correlation between the self-assembly of K7W6 and its membrane activity.

Molecular Mechanism Underlying the Assembly-Induced Function of K7W6. To gain insights into the underlying mechanism of this correlation, we conducted molecular dynamics (MD) simulations with the coarse-grained (CG) MARTINI force field to investigate the interactions between K7W6 monomers or clusters and lipid membranes.^{47,48} Our simulations validate the assembly-induced function of K7W6. As depicted in Figures 5a and S8, K7W6 clusters exhibit specific binding affinity toward PC/PG membranes rather than PC membranes, while K7W6 monomers nonspecifically embed themselves in both types of membranes. The computation of membrane binding free energy for K7W6 also supports this functionality: there are discernible differences in free energy changes when peptide

clusters bind to PC and PC/PG membranes; however, distinguishing changes in free energy for monomer–membrane binding is challenging in these two scenarios (Figure 5b). Further decomposition of the free energy into enthalpic and entropic contributions suggests that successful membrane binding of a K7W6 cluster is predominantly governed by an enthalpic change in the system (Figure 5c). Specifically, when K7W6 sequences self-assemble into a cluster, hydrophobic W residues are shielded by positively charged K residues to form an approximate core–shell structure that facilitates attraction between the K7W6 cluster and a PC/PG membrane, resulting in the enthalpy-driven process (Figures S9 and S10). Therefore, selective membrane-binding behavior exhibited by K7W6 clusters represents an assembly-induced function regulated by enthalpic change.

Interestingly, as demonstrated in the simulations, the K7W6 cluster interacts with PC/PG membranes through a two-step mechanism. Initially, it adheres to the membrane by embedding itself at the interface between phospholipid

headgroups and tails, as discussed above (Figure 5a). Subsequently, it disassembles stepwise (Figures 5a and S11). Notably, this behavior differs from that of AMPs like melittin, a natural amphipathic α -helical peptide, which typically exhibits strong aggregation on membranes.^{43,49} By calculating the disassembly free energy of a peptide cluster (where disassembly refers to one peptide detaching from the cluster), we discovered that on a membrane surface, this free energy reaches its minimum at a relatively large L value (i.e., separation distance between detached peptide and cluster), whereas in solution, this free energy dramatically increases with L (Figure 5d). Thus, this disassembly behavior is entirely mediated by the membrane. Furthermore, we found that an increase in entropy primarily leads to a decrease in the disassembly free energy (Figure 5e). Binding to the membrane results in the deformation of the cluster structure and exposure of hydrophobic W residues to the membrane surface (Figures S9), facilitating cluster disassembly. Additionally, dispersion of PG lipids occurs during cluster disassembly due to peptide adsorption-induced changes in the aggregation state (Figure S11). All these alterations contribute to an increase in entropy. Consequently, peptide clusters undergo an entropy-driven process when undergoing disassembly on membranes. Generally, the high anti-TB activity and efficiency for disrupting membranes induced by the assembly of K7W6 can be attributed to enthalpy-driven binding of clusters on specific membrane surfaces followed by entropy-driven monomerization for effective interaction with membranes.

CONCLUSIONS

Using an AI-assisted approach, we have successfully employed de novo design to develop a minimal model peptide against TB. The designed K7W6 sequence exhibits potent antimycobacterial activity against *M. smegmatis* mc²155, *M. tb* H37Rv, and *M. bovis* BCG as well as clinically isolated *M. tb* and rifampicin-resistant *M. tb* strains. Furthermore, K7W6 demonstrates excellent synergism with the first-line antibiotic rifampicin, low propensity to induce drug resistance, and moderate cytotoxicity toward mammalian cells such as RAW246.7 and Vero and effectively suppresses *M. tb* infection *in vivo*. These performances can be attributed to the highly selective and efficient membrane permeabilization activity of self-assembled K7W6 clusters, which has been confirmed through a combination of biophysical experiments and simulation approaches. This assembly-triggered membrane selectivity, coupled with enhanced activity, distinguishes it from most natural antimicrobial peptides that primarily exhibit concentration-dependent activity due to intermolecular cooperative effects during their interactions with cell membranes. Our findings unveil a complex enthalpy–entropy competition mechanism during the membrane interaction process of K7W6 clusters: initial selective binding to the membrane is controlled by enthalpy followed by entropy-driven disassembly on the membrane surface that maximizes the individual sequences' ability to disrupt membranes. This study provides valuable insights into rational peptide design for treating susceptible and drug-resistant TB infections by leveraging AI technology to enhance our understanding while uncovering distinct molecular mechanisms underlying the membrane activity induced by self-assembled K7W6 sequences.

MATERIALS AND METHODS

Materials. Phospholipids, including 1,2-dioleoyl-*sn*-glycero-3-phosphocholine (DOPC), 1,2-dioleoyl-*sn*-glycero-3-phospho-(1'-*rac*-glycerol) (sodium salt) (DOPG), and 1,2-dipalmitoyl-*sn*-glycero-3-phosphoethanolamine-*N*-(lissamine rhodamine B sulfonyl) (RhB-PE) were purchased from Avanti Polar Lipids and used as received. The peptides, from K13W0 to K2W11, with a purity >95% was obtained from Synpeptide Co., Ltd. (Nanjing). Calcein and phosphate buffer solution (PBS; with 8006 mg L⁻¹ NaCl, 201 mg L⁻¹ KCl, 272 mg L⁻¹ KH₂PO₄, and 2865 mg L⁻¹ Na₂HPO₄·12H₂O; sterilized through 0.1 μ m filtration, pH 7.2–7.4) were purchased from Sigma-Aldrich or Pricella (Wuhan). All the experiments were performed at the room temperature of 25 °C unless otherwise stated.

Bacterial Strains. *M. tuberculosis* H37Rv (ATCC 27294) and clinically isolated *M. tb* strains (*M. tuberculosis* WXY, *M. tuberculosis* CAS3, and rifampicin-resistant *M. tuberculosis* FYX) were obtained from the Affiliated Hospital of Zunyi Medical University or Kunming Medical University. *M. smegmatis* mc²155, *M. bovis* BCG and mCherry-*M. bovis* BCG were provided by the Jiangsu Provincial Key Laboratory of Infection and Immunity of Soochow University. *M. tuberculosis* H37Rv, *M. bovis* BCG, mCherry-*M. bovis* BCG, and clinically isolated *M. tb* were cultured in a Middlebrook 7H9 medium or enumerated on 7H11 agar supplemented with 10% oleic acid–albumin–dextrose–catalase, 0.50% glycerol, and 0.05% Tween 80. *M. smegmatis* mc²155 was cultured in a Luria–Bertani (LB) broth and enumerated on LB agar.

Giant Unilamellar Vesicle Leakage Assay. GUVs were prepared following the conventional electro-formation method.^{49,50} Briefly, a solution of lipids (PC or PC/PG, labeled with 0.5 wt % RhB–PE; 60 μ L $\times$ 2.0 mg mL⁻¹ in chloroform) was deposited onto two ITO-coated glass slides and dried under a vacuum overnight. The dry film was transferred to a homemade electro-formation chamber (with the two glass slides as electrodes) and rehydrated in 0.1 M sucrose buffer. Alternating voltages were applied (0.5 V $\times$ 20 min, 1.0 V $\times$ 20 min, and 1.5 V $\times$ 3 h). The obtained vesicles were washed three times *via* centrifugation (8000 rpm $\times$ 20 min). Well-dispersed GUVs with a size distribution of 8–30 μ m were collected, redispersed in 0.1 M sucrose buffer containing 0.2 mg mL⁻¹ calcein (with a final lipid concentration of $\sim$ 0.02 mg lipid mL⁻¹),⁴⁹ and transferred to a sample chamber for observation.

A homemade chamber cell with a cover-glass substrate was used for microscopy observation in the experiments. For the immobilization of the vesicles, the glass substrate was washed carefully, dipped into APTES for 5 min, and dried under N₂ flow. After that, the glass slide was kept at 120 °C for 30 min before being installed in the chamber for use. For the *in situ* microscopy observations, a volume of the GUV dispersion was transferred to the chamber cell and stabilized for about 5 min for particle immobilization.

After the immobilization of GUVs on the substrate, a certain amount of peptide solution (to a final concentration of 0.5, 1.0, 2.0, 5.0, or 10.0 μ g mL⁻¹ as specified in the main text) was added for *in situ* monitoring. Optical observations were performed on an inverted confocal laser scanning microscope (LSM 710, Zeiss) equipped with a 63 $\times$ oil objective. Signals from the RhB channel for lipids (EX 543 nm, EM 575–640 nm), calcein channel (EX 488 nm, EM 530/50 nm), and the overlaid channel were captured simultaneously. The data were acquired with a pixel dwell time of 1.58 μ s, and it generally took 3.87 s to take one image. Meanwhile, the transmission channel illuminated with a halogen lamp was acquired. All images were captured under the same instrumental settings. During the dynamic entry process of calcein into the interior of a GUV due to peptide exposure, the fluorescence intensity of the GUV interior at each time point (i.e., mean value among pixels read out directly from the Zeiss LSM software) was normalized with that of the surrounding environment and plotted as a function of time. More than five parallel experiments were performed in each condition, and the representative GUVs are shown in the main text.

Photovoltage Transient Assay. Setup of the System. The two electrodes, including a silicon wafer (6.5 cm $\times$ 0.8 cm) as the working

electrode and an Ag/AgCl reference electrode, were installed in a quartz cuvette containing 1.5 mL of DI water. Before installation, the silicon wafer was thoroughly cleaned, with the back and edges being sealed with insulating glue, and the front face being deposited with a PC or PC/PG bilayer by the traditional vesicle fusion method. A square-wave-modulated illumination (250 ms ON and 500 ms OFF) was generated by a xenon lamp (500 W) and produced on the silicon wafer as the input signal. The output voltage signal was collected simultaneously.

Data Collection. A certain amount of peptide was added to the solution to a certain concentration as specified in the main text, and the photovoltage tests were began after stabilization of the system. The time point of peptide adding was set as $t = 0$. The data were collected per minute (a number of three square-wave cycles were carried out each time for average) until the output data became constant, based on which a $\tau \sim t$ profile was acquired. Tests on membrane exposed to pure water or peptides with varying concentrations were carried out respectively, each repeated for at least five times.

Determination of τ . For each photovoltage pulse, the charge relaxation time constant, τ , was obtained through fitting the descending edge with $y = C_1 e^{-t/\tau} + C_2 e^{-t/\tau'} + C_3$, in which C_1 , C_2 , and C_3 are constants. Here, τ represents the diffusion of electrons into the solution (i.e., the relaxation time) and τ' represents the diffusion of electrons on the surface of the silicon wafer, which is about 50 μs .⁵¹ For clarity, the initial 70% of the descending edge is used for each pulse in this work.⁴⁶

Minimal Inhibitory Concentration Determination. A 2-fold broth microdilution method was used to determine the antimicrobial activity of peptides following the standard protocols.⁵² Briefly, a 2-fold serial dilution of the peptide was prepared in a 96-well plate (100 μL /well). Mycobacteria were suspended in Middlebrook 7H9 broth (approximately 10^5 CFU/mL), added to the plate (100 μL /well), and cultured at 37 °C after 7 days for *M. tuberculosis* and 72 h for *M. smegmatis*. At the end of incubation, the absorbance at 600 nm of the microbial inoculums was recorded by a microplate reader (Bio-Tek ELX800, U.S.A.). The minimal concentrations at which no mycobacterial growth occurred was recorded as MIC values. The traditional antibiotic rifampicin was used as the positive control.

Mycobacterial Killing Kinetic Assay. Tubes containing 10 mL of Middlebrook 7H9 broth with the peptide at concentrations corresponding to 1 $\times$, 2 $\times$, and 4 $\times$ MIC were inoculated with mycobacteria at a density of approximately 5.5×10^5 CFU/mL and incubated at 37 °C. Viable counts of the mycobacteria were tested after 8, 24, 48, and 72 h for *M. smegmatis* and 4, 7, 14, and 21 days for *M. tuberculosis* H37Rv, respectively.⁵³

SEM Specimen Preparation and Imaging. After being incubated with the peptide at 1 $\times$ MIC for 4 h, the mycobacteria were harvested, centrifugated at 1000g for 10 min, fixed in 2.5% glutaraldehyde at 4 °C for 2 h, and postfixed in 1% osmium tetroxide for 2 h. Then, the mycobacteria were dehydrated in a series of gradients of ethanol (30, 50, 70, 90, and 100%), frozen in liquid nitrogen-cooled *tert*-butyl alcohol, and vacuum-dried overnight. Finally, the cells were mounted onto aluminum stubs and vacuum sputter-coated with gold. The outer morphology of the mycobacteria was observed using a Hitachi S-4800 scanning electron microscope.

Cellular Viability Assays. RAW246.7 and Vero cells were obtained from National Collection of Authenticated Cell Cultures (<https://www.cellbank.org.cn/>) and were cultured in Dulbecco's modified Eagle's medium supplemented with antibiotics and 10% fetal bovine serum. The cells were seeded in 96-well plates at a density of 2×10^4 cells per well. A 2-fold serial dilution of the peptide was subsequently added to each well and incubated at 37 °C for 24 h. Following this, CCK-8 solution (10 μL per well) was introduced and incubated for an additional 2 h, with absorbance at 450 nm being monitored.⁵⁴

Potential of Combining K7W6 with Rifampicin. Checkerboard assay was performed to investigate the synergistic actions of K7W6 and rifampicin. In brief, a series of 2-fold K7W6 or rifampicin

dilutions were prepared. After K7W6 (50 μL /well) and rifampicin (50 μL /well) were mixed in a 96-well plate, *M. smegmatis* or *M. tuberculosis* H37Rv (10^5 CFU/mL, 100 μL /well) was added to the K7W6-rifampicin mixture. The plate was cultured at 37 °C and observed visually or recorded spectrophotometrically at 600 nm after 7 days for *M. tuberculosis* and 72 h for *M. smegmatis*.⁵⁴

For drug resistance stimulation, *M. smegmatis* and *M. tuberculosis* H37Rv were repeatedly treated with rifampicin, K7W6, or vehicle (DMSO, PBS) for 10 passages. For every passage, *M. smegmatis* and *M. tuberculosis* H37Rv were incubated with a sub-MIC concentration of rifampicin or K7W6 (1/8 MIC at that particular passage) until growing to the log phase, and then the MICs of K7W6 and rifampicin against *M. smegmatis* and *M. tuberculosis* H37Rv were tested.⁵⁴

To investigate the potential of the combined use of K7W6 with rifampicin, we evaluated whether K7W6 can inhibit rifampicin-induced drug resistance in *M. smegmatis* and *M. tuberculosis* H37Rv. Briefly, *M. smegmatis* and *M. tuberculosis* H37Rv were passaged for 10 times in the presence of rifampicin and K7W6. For each passage, *M. smegmatis* or *M. tuberculosis* H37Rv was exposed to a sub-MIC concentration of rifampicin (1/8 MIC at that particular passage) and a constant concentration of K7W6 (1/4 MIC of K7W6 against *M. smegmatis* or *M. tuberculosis* H37Rv) until growing to the log phase. The MIC value of rifampicin against *M. smegmatis* or *M. tuberculosis* H37Rv was tested then.⁵⁵

Antimycobacterial Activity in Mice. Female C57BL/6J mice ($n = 6$, 18–20 g) were intranasally inoculated with *M. smegmatis* (2×10^7 CFU/mouse), *M. bovis* BCG (1.5×10^7 CFU/mouse), or mCherry expressing *M. bovis* BCG (mCherry-BCG, 3×10^7 CFU/mouse). 1 h after *M. tb* infection, K7W6 (5 mg kg⁻¹) or PBS (solvent of K7W6) was delivered into mouse airways with a pipette tip. 1 week post *M. smegmatis* inoculation and 4 weeks post *M. bovis* BCG inoculation, the lungs were taken for mycobacterial load and histopathology assay after sacrifice. For mycobacterial load assay, the left lungs were homogenized in PBS (1 mg of lung/mL of PBS) on ice. A series of diluted homogenates (10^{-1} , 10^{-2} , 10^{-3} , 10^{-4} , 10^{-5} , and 10^{-6}) were plated onto Middlebrook 7H11 agar plates, and CFUs of *M. smegmatis* and *M. bovis* BCG were counted after culture at 37 °C for 72 h and 4 weeks, respectively. For histopathological study, the right lungs were taken and fixed with 10% formalin for 24 h. After dehydrating by an increasing gradient of alcohol, the lungs were embedded in paraffin, cut into sections (5 μm), stained with hematoxylin and eosin, and observed using a light microscope. 4 weeks post mCherry-BCG inoculation, the lungs were taken to test the fluorescence intensity using an *in vivo* imaging system (IVIS SPECTRUM, PerkinElmer).⁵⁶

Simulation Methods. All MD simulations were performed by using the GROMACS 5.1.4 package with a coarse-grained MARTINI force field.^{47,48} The system temperature was maintained with the Berendsen temperature⁵⁷ coupling scheme with a time constant of 1 ps, and the pressure was controlled with the Parrinello–Rahman semiisotropic barostat⁵⁸ with a time constant of 12 ps and a compressibility of 4.5×10^{-5} bar⁻¹. All simulations were carried out in the isothermal–isobaric (NPT) ensemble at a temperature of 310 K and pressure of 1.0 bar. Nonbonded Lennard-Jones interactions were cut off at 1.1 nm, and electrostatic interactions were calculated with the PME method. Periodic boundary conditions were applied in all three directions.

Simulation Models and Protocol. Lipid bilayer membranes composed of DOPC or DOPC/DOPG = 9:1 were built using the script insane.py.⁵⁹ The initial membrane size was 20×20 nm². Peptides were built directly by Pymol software based on their amino acid sequence, which then were coarse-grained using the Martinize script with the MARTINI force field. Moreover, the peptide cluster was obtained by the self-assembly of peptides in solution; and the α -helical peptide structure was maintained in the simulations.

Peptide monomer or cluster was initially placed above the surface of the membrane (~ 2.0 nm), followed by energy minimization and system equilibrium. Then, the system was simulated for at least 5 μs with a time step of 20 fs. To ensure computational convergence, three independent runs were performed in each simulation condition.

Calculations of Free Energy, Entropy, and Enthalpy. The free energy change (ΔG) in the peptide–membrane interaction was evaluated from the potential of mean force (PMF) as a function of the peptide–membrane or interpeptide separation distance with an umbrella sampling method. Moreover, ΔG can be decomposed into entropy and enthalpy components *via* different temperature dependencies, as $-T\Delta S = T \frac{dG}{dT} \approx \frac{T}{2\Delta T}(G(T + \Delta T) - G(T - \Delta T))$, and $\Delta H = \Delta G - (-T\Delta S)$.^{60–62} The umbrella sampling calculations were carried out at three different temperatures (e.g., 300, 310, and 320 K). The entropy and enthalpy contributions were calculated by the above equations.

Machine-Learning Models. The MLP model, with its sequential layer structure and ReLU activation functions, served as a fundamental model for our antimicrobial peptide classification task. A train/validation/test split ratio of 75:15:10 for modeling was applied. The training and evaluation processes were designed to optimize the model's ability to distinguish peptides based on their antimicrobial properties. Our model consisted of input layer, first hidden layer, second hidden layer, and output layer. Due to the limited data size, transfer learning played a crucial role in our study. The above trained MLP model was used as the pretrained model. In our transfer learning strategy, we kept the parameters of the first layer and the output layer frozen and focused on fine-tuning the second layer of the MLP. The fine-tuned model was then trained on the smaller target data set related to *M. tb*.

SHAP Analysis. SHAP analysis provides an interpretive framework to understand the contribution of each variable in a machine learning model's prediction. In our case, SHAP analysis was employed to discern the influence of various physicochemical properties on the efficacy of antimicrobial peptides against specific bacterial strains. For SHAP analysis, we used models trained on the outer loop training sets, with hyperparameters identical to those of the final predictive model. The SHAP values were calculated using the test sets. This involved determining the contributions of each variable in a given sample from the test set to the model's prediction. We computed SHAP values using the KernelExplainer function of the SHAP Python package (version 0.41.029), by setting the number of sampling times at 1000. The KernelSHAP method utilizes a local interpretable model-agnostic explanations (LIME) strategy to calculate variable contributions (SHAP values).

Statistical Analysis. All simulation and experimental measurements were performed with at least three to five replicates for each condition/data point. Analysis of variance was performed using OriginPro 9.0 (OriginLab software, Northampton, MA), and the values are displayed as mean standard deviation. *P*-Values were calculated by ANOVA. **P* < 0.05, ***P* < 0.01, and ****P* < 0.001. Data set of experiment was performed with samples compared to controls unless otherwise stated.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsnano.4c05265>.

More details of machine learning; extracellular ATP test; more characterizations of the peptide assemblies; more data on the GUV leakage assays, photovoltage tests, and MD simulations; descriptors used in machine learning; and MIC determination and drug-resistance development of K7W6 (PDF)

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Author Contributions

*L.W., W.T., Y.X., and C.X. contributed equally. K.Y. and B.Y. conceived, designed, and supervised the project. L.W. synthesized the peptides and conducted live cell and *in vivo* assays. W.T. performed the simulations. Y.X. conducted machine learning and related analysis tasks. C.X. carried out the characterizations. Y.D. built the photovoltage equipment and collaborated with Y.G. to perform the photovoltage experiments. S.S. and Y.W. performed GUV leakage assays. K.Y., B.Y., and L.W. wrote the initial draft. K.Y. and B.Y. modified and finalized the manuscript. All authors approved the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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